INTRODUCTION TO APPLIED NUMERICAL METHODS

Case Study: Linear Equations

Lang Yuan, Associate Professor

UNIVERSITY OF South Carolina

College of Engineering and Computing

Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

$$[A]: Row(2) - Row(1) \times \left(\frac{a_{21}}{a_{11}}\right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Define: $f_{21} = \left(\frac{a_{21}}{a_{11}}\right)$

$$[A]: Row(3) - Row(1) \times \left(\frac{a_{31}}{a_{11}}\right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

Define: $f_{31} = \left(\frac{a_{31}}{a_{11}}\right)$

Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$[A]: Row(3') - Row(2') \times \left(\frac{a'_{32}}{a'_{22}}\right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

Define: $f_{32} = \left(\frac{a'_{32}}{a'_{22}}\right)$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}$$

$$[U] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Process

- **Step 2(i):** Forward substitution step to obtain {D}

$$[L]\{D\} = \{B\}$$

$$d_i = b_i - \sum_{j=1}^{i-1} l_{ij} d_j \quad \text{for } i = 2, 3, \ldots, n$$

$$\begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

- **Step 2(ii):** Back-substitution step to obtain {x}

$$[U]\{X\} - \{D\} = 0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix}$$

$$x_n = \frac{d_n}{u_{nn}}$$

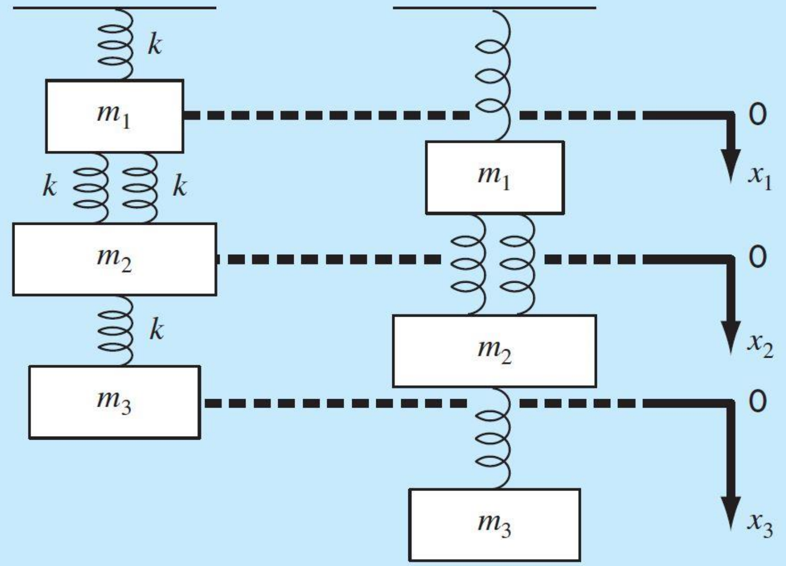
$$x_i = \frac{d_i - \sum_{j=i+1}^{n} u_{ij} d_j}{u_{ii}} \quad \text{for } i = n-1, n-2, \ldots, 1$$

MATRIX INVERSION

- Inversion of $[A]$: $[A][A]^{-1} = [I]$
- The inversion can be calculated in a column-by-column fashion by generating solutions with unit vectors as the right-hand-side constants.
- Example:
  - $[A] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
  - $[A]^{-1} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$
  - $[A][A]^{-1} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
  - Calculated the matrix in a column-by-column fashion (solving two linear equations):
  - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
  - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

SPRING-MASS SYSTEMS

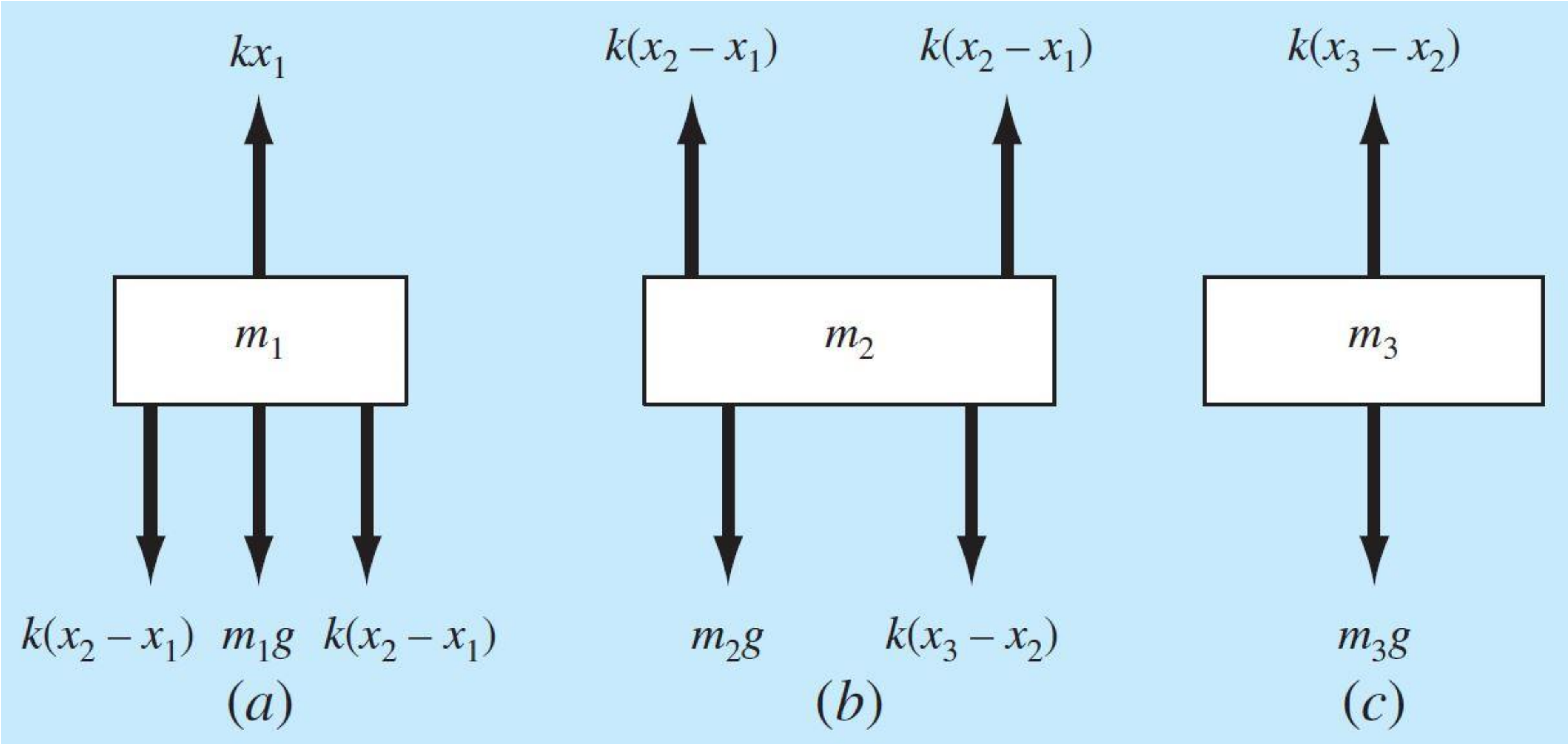
- A system composed of three masses suspended vertically by a series of springs. Please calculate the displacement of each mass, given:
  - Mass: $m_1 = 2\ kg, m_2 = 2kg, m_3 = 2.5kg$
  - Spring stiffness, $k = 10\frac{N}{m}$
  - Gravitational acceleration, $g = \frac{9.81m}{s^2}$
- Note that the positions of the masses are referenced to local coordinates with origins at their position before release.



(a) Before release

(b) After release.

FREE-BODY DIAGRAMS



STEADY-STATE

Set second derivatives to zero and collect terms:

$$\begin{aligned} 3kx_1 - 2kx_2 \qquad\quad &= m_1 g \\ -2kx_1 + 3kx_2 - \ kx_3 &= m_2 g \\ -\ kx_2 \ + kx_3 &= m_3 g \end{aligned}$$

or in matrix form:

$$[K]\{X\}=\{W\}$$

where {X} = vector of unknowns, {W} = vector of weights, and [K] is called the *stiffness matrix*

$$[K] = \begin{bmatrix} 3k & -2k & \\ -2k & 3k & -k \\ & -k & k \end{bmatrix}$$

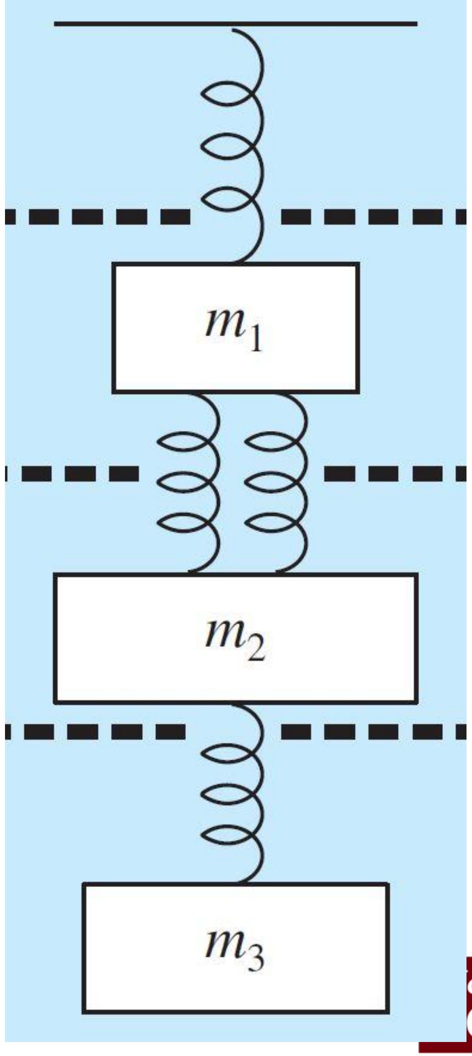
SOLUTION

Substitute parameter values:

$$\begin{bmatrix} 30 & -20 & \\ -20 & 30 & -10 \\ & -10 & 10 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 19.62 \\ 29.43 \\ 24.525 \end{Bmatrix}$$

and solve for the displacements:

$x_1 = 7.36$
$x_2 = 10.06$
$x_3 = 12.51.$



LU DECOMPOSITION

[*A*]{*x*}={*b*} [*L*][*U*]=[*A*]

Step 1: [*L*]{*d*}={*b*} is used to generate an intermediate vector {*d*} by forward substitution.
Step 2: [*U*]{*x*}={*d*} is used to get the answer, {*x*}, by back substitution.

$$[A] = \begin{bmatrix} 30 & -20 & \\ -20 & 30 & -10 \\ & -10 & 10 \end{bmatrix}$$

Elimination process:

$Row(2') = Row(2) - Row(1) * f_{21}$:

$$f_{21} = -\frac{20}{30} = -0.6667$$

$$\begin{bmatrix} 30 & -20 & \\ 0 & 30 - 20 * 0.6667 & -10 \\ & -10 & 10 \end{bmatrix} \longrightarrow \begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & -10 & 10 \end{bmatrix}$$

LU DECOMPOSITION

$$\begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & -10 & 10 \end{bmatrix}$$

Elimination process:

$Row(3') = Row(3) - Row(1) * f_{31}$:

$$\begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & -10 & 10 \end{bmatrix} \qquad f_{31} = 0$$

$Row(3'') = Row(3') - Row(2') * f_{32}$:

$$f_{32} = -\frac{10}{16.6660} = -0.6$$

$$\begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & 0 & 10 - 10 * 0.6 \end{bmatrix}$$

$$[U] = \begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & 0 & 4 \end{bmatrix}$$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}$$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ -0.6667 & 1 & 0 \\ 0 & -0.6 & 1 \end{bmatrix}$$

LU DECOMPOSITION

$$\begin{bmatrix} 30 & -20 & \\ -20 & 30 & -10 \\ & -10 & 10 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 19.62 \\ 29.43 \\ 24.525 \end{Bmatrix}$$

[*A*]{*x*}={*b*} [*L*][*U*]=[*A*]

Step 1: [*L*]{*d*}={*b*} is used to generate an intermediate vector {*d*} by forward substitution.
Step 2: [*U*]{*x*}={*d*} is used to get the answer, {*x*}, by back substitution.

$$[A] = \begin{bmatrix} 30 & -20 & \\ -20 & 30 & -10 \\ & -10 & 10 \end{bmatrix} \qquad [L] = \begin{bmatrix} 1 & 0 & 0 \\ -0.6667 & 1 & 0 \\ 0 & -0.6 & 1 \end{bmatrix} \qquad [U] = \begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & 0 & 4 \end{bmatrix}$$

A forward-substitution to compute {D}: $\boxed{[L]\{D\} = \{B\}}$

$$\begin{bmatrix} 1 & 0 & 0 \\ -0.6667 & 1 & 0 \\ 0 & -0.6 & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 19.62 \\ 29.43 \\ 24.525 \end{Bmatrix} \Rightarrow \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 19.62 \\ 42.5107 \\ 50.0314 \end{Bmatrix}$$

$$\begin{bmatrix} 30 & -20 & \\ -20 & 30 & -10 \\ & -10 & -10 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 19.62 \\ 29.43 \\ 24.525 \end{Bmatrix}$$

LU DECOMPOSITION

[*A*]{*x*}={*b*}

[*L*][*U*]=[*A*]

Step 1: [*L*]{*d*}={*b*} is used to generate an intermediate vector {*d*} by forward substitution.
Step 2: [*U*]{*x*}={*d*} is used to get the answer, {*x*}, by back substitution.

$$[A] = \begin{bmatrix} 30 & -20 & \\ -20 & 30 & -10 \\ & -10 & -10 \end{bmatrix} \quad [L] = \begin{bmatrix} 1 & 0 & 0 \\ -0.6667 & 1 & 0 \\ 0 & -0.6 & 1 \end{bmatrix} \quad [U] = \begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & 0 & -16 \end{bmatrix}$$

A back-substitution to compute {X}:

$$\boxed{[U]\{X\} = \{D\}}$$

$$\begin{bmatrix} 30 & -20 & \\ 0 & 16.6660 & -10 \\ & 0 & 4 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 19.62 \\ 42.5107 \\ 50.0314 \end{Bmatrix} \Rightarrow \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.3578 \\ 10.0558 \\ 12.5078 \end{Bmatrix}$$

OVERVIEW

- Significant figures
- True fractional relative error, approximated relative percent error, pre-specified percent error
- The form of Taylor series
- Numerical differentiation: forward, backward, centered
- Conditional number
- Bisection method: $x_l, x_u, x_r, f(x_l), f(x_u), f(x_r), \varepsilon_a, \varepsilon_s$
- Newton Raphson method: $x_i, f(x_i), f'(x_i), x_{i+1}\,,, \varepsilon_a, \varepsilon_s$
- Basics of matrix: row, column, operations
- Gauss elimination: forward elimination, backward substitution
- LU decomposition: forward elimination for [L] and [U], forward substation for intermediate vector {d}, backward substation for {x}

Engineering and Computing